

PRACTICAL WEEK-3
2520090159
A SANJANA REDDY

The image displays two screenshots of the MongoDB Data Explorer interface, showing the 'book_metadata' collection in the 'bookflow' database. The interface includes a sidebar with a cluster tree, a main panel for document viewing, and a bottom status bar.

Top Screenshot:

- Cluster:** Cluster0
- Database:** bookflow
- Collection:** book_metadata
- Documents:** 1
- Query:** Type a query: { field: 'value' } or [Generate query](#)
- Document 1:**

```
{
  "_id": ObjectId('6a91432f6d442b6cd5cf3e80'),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3)
}
```

Bottom Screenshot:

- Cluster:** Cluster0
- Database:** bookflow
- Collection:** book_metadata
- Documents:** 5
- Query:** Type a query: { field: 'value' } or [Generate query](#)
- Document 1:**

```
{
  "_id": ObjectId('6a9143ae6d442b6cd5cf3e86'),
  "book_id": 105,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "5.0 MB",
  "member_id": 105,
  "reviews": Array (2)
}
```

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Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

🔍

{

"reviews.rating": {

"\$gt": 4

}

}

🔍

Explain

Reset

Find

Options

25

1 - 5 of 5

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⌲

member_id: 101

rating: 5

comments: "Excellent science fiction story"

1: Object (3)

member_id: 102

rating: 4

comments: "Very interesting and informative"

2: Object (3)

member_id: 103

rating: 5

comments: "Excellent space adventure"

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Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

🔍

{

"reviews.rating": {

"\$in": [4, 5]

}

}

🔍

Explain

Reset

Find

Options

25

1 - 5 of 5

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⌶

⌲

0: Object (3)

member_id: 101

rating: 5

comments: "Excellent science fiction story"

1: Object (3)

member_id: 102

rating: 4

comments: "Very interesting and informative"

2: Object (3)

member_id: 103

rating: 5

comments: "Excellent space adventure"

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - local
 - studentDB
 - students

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

{
 "format": {
 "\$regex": "Digital",
 "\$options": "i"
 }
}

25 1-4 of 4

_id: ObjectId('6a91432f6d442b6cd5cf3e88')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - local
 - studentDB
 - students

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

Untitled - modified + CREATE NEW

PREVIEW

Stage 1 \$match

1 {
2 "published_year": {
3 "\$gt": 2020
4 }
5 }

Output preview after \$match stage (Sample of 4 documents)

_id: ObjectId('6a91432f6d442b6cd5cf3e88')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

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Cluster0

book_metadatadata

+

Cluster0 > bookflow > book_metadatadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

+

?

Explain

Run

Options

Untitled - modified

+ CREATE NEW

PREVIEW

{}

 </>

⚙

Stage 2 \$unwind

1 /**
2 * path: Path to the array field.
3 * includeArrayIndex: Optional name for
4 * preserveNullAndEmptyArrays: Optional
5 * toggle to unwind null and empty val
6 */
7 {
8 path: "\$reviews"
9
10 }

Output preview after \$unwind stage (Sample of 10 documents)

```
_id: ObjectId('6a9143ae6d442b6cd5cf3e86')
book_id: 185
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 185
```

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Cluster0

book_metadatadata

+

Cluster0 > bookflow > book_metadatadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

\$group

+

?

Explain

Run

Options

Untitled - modified

+ CREATE NEW

PREVIEW

{}

 </>

⚙

Stage 3 \$group

1 {
2 "_id": "\$title",
3 "average_rating": {
4 "\$avg": "\$reviews.rating"
5 },
6 "total_reviews": {
7 "\$sum": 1
8 }
9 }
10 }

Output preview after \$group stage (Sample of 4 documents)

```
_id: "Data Structures"
average_rating: 4
total_reviews: 2
```

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The screenshot displays the MongoDB Atlas Data Explorer interface. On the left, the 'Data Explorer' sidebar shows the database structure: Cluster0 > bookflow > book_metadata. The main panel is titled 'Aggregations' and shows a pipeline with four stages: \$match, \$unwind, \$group, and \$sort. The \$sort stage is selected, and the output preview shows a sample of 4 documents with fields _id, average_rating, and total_reviews. The output preview shows a sample of 4 documents with fields _id, average_rating, and total_reviews.

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - local
- studentDB
 - students

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 2Validation

{ "\$text": { "\$search": "excellent" } }

251-5 of 5

Explain

Reset

Find

Options

_id: ObjectId('6a91432f6d442b6cd5cf3e80')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

reviews: Array (3)

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ORGANIZATIONPROJECTIXSCAN1/1

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree

Raw Output

"parsedTextQuery": {
 "terms": ["excel"],
 "negatedTerms": [],
 "phrases": [],
 "negatedPhrases": []
},
"textIndexVersion": 3,
"inputStage": {
 "stage": "FETCH",
 "inputStage": {
 "stage": "IXSCAN",
 "keyPattern": {
 "_fts": "text",
 "_ftsx": 1
 },
 "indexName": "reviews.comments_text",
 "isMultiKey": true,
 "isUnique": false,
 "isSparse": false,
 "isPartial": false,
 "indexVersion": 2,
 "key": {}
 }
}

Query Performance Summary

5 documents returned

5 documents examined

0 ms execution time

Is not sorted in memory

5 index keys examined

Query used the following index:
_fts (text)
_ftsx ↑

Close

